
CHEMWORLD: TASK SUCCESS DOES NOT ENSURE SCIENTIFIC GENERALIZATION

Anonymous authors

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ABSTRACT

Evaluating an autonomous scientist requires more than measuring the best experiment it finds. Its chosen observations must also support predictions of new interventions. We introduce ChemWorld, a programmable framework that separates private process laws, public experimental interfaces, and supplied information while preserving persistent state, finite resources, and exact environment replay. We study 240 autonomous research campaigns across six chemical system families with system-specific goals and Opaque, Aligned, and MisIndexed information conditions. Operational and predictive outcomes diverge: electrochemical optimization improves independently retested recommendations in 26 of 30 matched comparisons, but prediction in only 14. Doubling research budgets improves electrochemical and partitioning prediction without uniformly resolving crystallization errors. Information effects are also conditional. In aqueous equilibrium, Aligned improves prediction on nine questions but worsens it on the three most dilute conditions in all five worlds. Conversely, crystallization purity forecasts lose to the mean of their own observations in all thirty campaigns, while recovery forecasts beat it in twenty-six. These complementary patterns motivate evaluating whether agents know when an observed relationship should persist or change. Complete outcome records, response-specific metrics, and selected trajectory analyses support this diagnosis without equating successful operation, plausible explanation, or a single prediction score with scientific understanding.

1 INTRODUCTION

An autonomous scientist must decide both what to do and what its observations justify. A productive recipe is valuable, but need not support accurate predictions after changing material, process history, or operating regime. Conversely, an immediately unproductive experiment may distinguish explanations that matter for later inference. Task achievement and scientific generalization are therefore connected capabilities whose relationship must be measured, rather than assumed.

Language-model agents can plan chemical experiments, use specialist tools, and interact with automated laboratories (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023). Evaluating the resulting research is difficult because prior information influences which experiments are selected, and those experiments influence the subsequent account. Physical mechanisms are rarely fully known, while rerunning a campaign under matched starting conditions is expensive. A final optimization score cannot determine whether an agent acquired discriminating evidence or made warranted use of it.

We introduce **ChemWorld**, a composable experimental environment with separately controlled process laws, observations, and public information. Agents act through laboratory operations with persistent physical consequences and finite resources. The evaluator can reconstruct the exact evidence available before a scientific claim or prediction. Our protocol retains autonomous experiment selection and unrestricted mechanism expression, then measures operating recommendations, withheld-condition predictions, and uncertainty separately.

Across 240 campaigns in six system families, we find a selective separation between operational achievement and predictive knowledge. Additional resources and informative dossiers help in several settings, but neither produces a uniform improvement across responses. A finer analysis reveals complementary errors: equilibrium predictions can preserve an observed plateau outside its applicable

regime, whereas crystallization predictions can depart unnecessarily from stable observed purity. These are behavioural findings from one agent configuration, not a universal deficit or an identified psychological mechanism.

Our contributions are (i) a programmable instrument with internal qualification and exact environment replay, (ii) an autonomous research protocol comparing matched information conditions without prescribing the form of the mechanism, and (iii) complete selected cohorts and response-specific analyses showing why task scores alone are insufficient for evaluating scientific generalization. The platform spans nine task families; the present agent evidence covers six. Negative results and source-budget shortfalls remain visible.

2 CHEMWORLD AND THE RESEARCH PROTOCOL

2.1 PHYSICAL CONTROL WITH RESTRICTED OBSERVATIONS

A world comprises a public experimental contract and private process parameters, $\mathcal{W} = (W_{\text{pub}}, \theta)$. Compatible process components declare states, parameter domains, dependencies, and interfaces; operation and observation models turn these declarations into an executable laboratory. Chemistry components include reaction, thermal, phase, separation, crystallization, distillation, flow, and electrochemical processes. Task families can share components and are not independent laws.

Agents access declared operations and measurements, not θ , simulator code, or evaluator records. Sampling can consume material, separation changes its location, and thermal history persists. A validated transition commits to the state; a rejected action retains the committed state and records its declared attempt consequences. Replay reconstructs the environment and executed actions, including rejection and rollback. It does not imply deterministic model responses.

The evaluator can change a private law while preserving an interface, or intervene on public information while holding physics fixed. The present three-arm study uses the latter control. Platform qualification passes 64 task-world units, 1,786 boundary and categorical recipes, 192 invalid-action probes, and 52 generated compositions. The latter include 18 new topologies, eight new task-world identities on a registered topology, and 26 coverage cases. Module and interface tests cover 32 probes, seven valid paths, and seven invalid compositions. These are finite-domain internal checks, not wet-laboratory validation or agent discoveries.

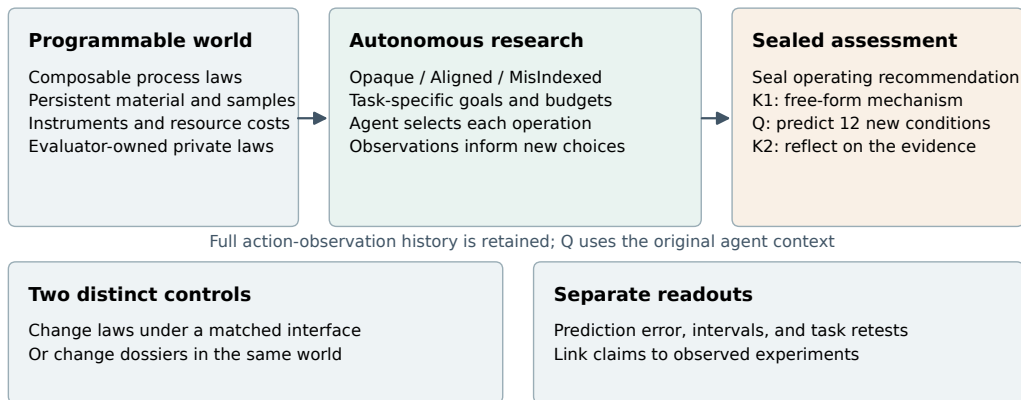


Figure 1: World control and assessment chronology. Agents freely choose experiments and mechanisms within the public contract. An operating recommendation is sealed before K1 and independently retested. Q receives the full experimental history; K2 follows sealed predictions. The protocol therefore does not impose report-only knowledge transfer.

2.2 SYSTEM-SPECIFIC COMMISSIONS AND THREE INFORMATION ARMS

Every study block has five worlds and three matched arms. **Opaque** exposes usable contracts and anonymous material identifiers without the tested instance-specific dossier. **Aligned** adds the nominal information associated with the intended entities, effective parameters, or process relationships. **MisIndexed** applies the block’s fixed mismatch or reassignment. Aligned is partial information rather than complete simulator truth; Opaque does not remove general chemical knowledge. The treatment locus—entity (E), parameter (P), or structure (S)—is distinct from the three arms.

Electrochemistry (EC) and reaction/thermal processing (RX) separately assign discovery and public-score optimization. Phase partitioning (PA) and aqueous equilibrium (EQ) emphasize characterization. Crystallization (C) seeks seed-excluded recovery subject to purity and fines constraints. Purification seeks original-charge recovery subject to purity. The latter system abbreviation P is distinct from the parameter-prior locus. Tasks are not forced into one yield objective.

The matrix contains EC/E 60 campaigns, PA/E 30, RX/P and RX/S 60, EQ/E, EQ/P, and EQ/S fifteen each, C/E 30, and purification/E 15. EC, PA, and C compare 12 and 24 batches; other blocks use twelve. All campaigns use GPT-5.6 Sol with medium reasoning effort in independent sessions. A budget supplies the same number of optional additional measurements and a final assay per completed batch. Other operational, material, time, and computation limits are explicit. Agents choose repetitions, measurement allocation, and grouping; a batch is not necessarily a separate model response or feedback-conditioned decision. No fixed equation family or per-batch reflection rule is imposed.

2.3 SEPARATE ASSESSMENTS OF ACHIEVEMENT, PREDICTION, AND EXPLANATION

For EC, RX, C, and purification, a completed-batch recommendation is sealed before assessment and its procedure independently retested in the same world using a different observation seed. Each campaign then produces **K1**, a free-form mechanistic account; **Q**, point predictions and intervals for twelve fixed new-batch recipes without laboratory access or target feedback; and **K2**, retrospective questions about evidence, alternatives, and discriminating experiments. An additional equilibrium parameter supplement yields fifteen submissions. The original research context is retained throughout. Thus Q can use more than K1, while K2 cannot change Q or establish what the agent believed earlier.

For campaign c and response m , error is

$$\text{MAE}_{cm} = \frac{1}{12} \sum_{q=1}^{12} |\hat{y}_{cqmq} - y_{qm}^{\text{ref}}|.$$

Coverage and interval width are reported alongside MAE. PA uses nominal 90% intervals; others use 80%. EC and purification use fixed seeded reference observations, PA and C underlying pre-assay values, and RX/EQ five-repeat means for point error with individual repeats for coverage. Macro error averages responses within the specified block. Different scales and reference targets preclude a pooled cross-system accuracy ranking.

Goal pairs match world, arm, budget, and locus; budget pairs compare independent sessions matched on the other factors. The larger budget includes greater computation and operation allowances, so it is a resource-envelope intervention. There is one realization per cell and five reused world clusters per study. We report descriptive magnitudes and paired directions without counting correlated questions, responses, or arms as independent replicates.

All 720 canonical assessments and 165 retests completed. One C source has eleven assays after solvent exhaustion; one purification source has ten assays and two discards from twelve starts. The pool therefore has 238 conforming chains and 3,597/3,600 planned source assays. Both shortfalls are retained. Infrastructure attempts are accounted for separately, and source outcomes are not replaced for being unfavourable. These agent cohorts retain their development-evidence status, distinct from frozen platform qualification.

3 OPERATIONAL SUCCESS DOES NOT ENSURE BETTER PREDICTION

EC optimization improves recommendation retests in **26/30** pairs, but score-prediction error in **14/30**. Thirteen pairs combine improved retests with worse prediction, distributed across all five worlds (Figure 2). Of the remaining pairs, thirteen improve both endpoints. Restricting to the seventeen pairs without infrastructure-driven source restarts retains fifteen retest improvements, eight prediction improvements, and seven discordances. The association therefore does not arise solely from the retained restart asymmetry.

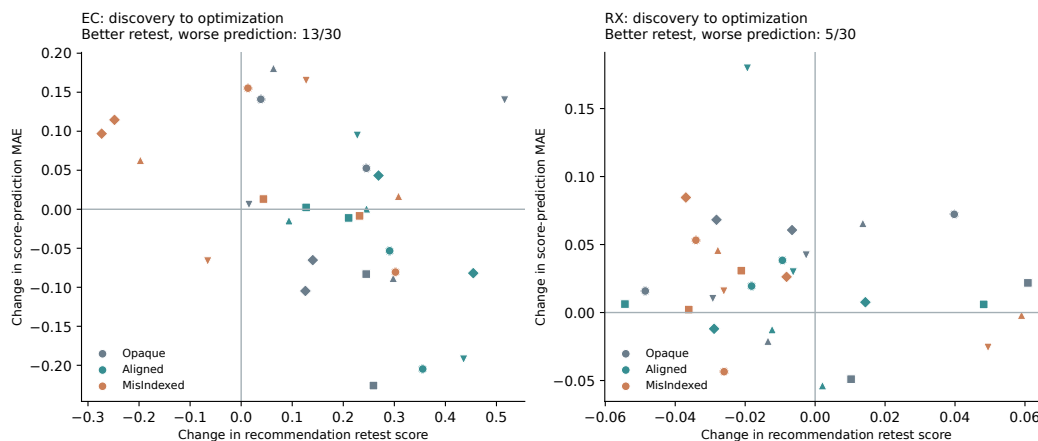


Figure 2: Matched optimization versus discovery outcomes. EC shows frequent improvement in recommendation performance without better score prediction. RX supplies a contrasting system in which the optimization commission itself often fails to improve retests. Each point represents a matched pair, with world reuse across arms and budgets; scales are specific to each system.

RX gives a different pattern: optimization improves retests in 9/30 pairs, score prediction in 8/30, and the original six-response macro error in 6/30. The optimization instruction is therefore not an oracle for finding better experiments. These contrasts concern complete research policies; they do not causally separate experimental acquisition from subsequent interpretation.

Constrained delivery adds another distinction. Purification recommendations are purity-eligible in 3/5 Aligned worlds and 0/5 in each other arm, although mean recovery is lowest for Aligned: 0.04455 versus 0.12373 for Opaque and 0.32105 for MisIndexed. All procedures are legal. A high recovered fraction can therefore coexist with failure to deliver the requested quality. One Opaque retest reaches purity 0.7992 against a 0.80 threshold, also illustrating why one noisy retest should not be equated with an exact reliability estimate.

4 LARGER BUDGETS HELP, BUT DO NOT RESOLVE EVERY RESPONSE

The 24-batch condition reduces EC mean score MAE from 0.17378 to 0.11222 under discovery and from 0.17811 to 0.10818 under optimization. Eleven and twelve of the respective fifteen pairs improve. Aggregate error decreases on all six EC responses, with mean directions preserved under leave-one-world-out checks. Prediction is therefore sensitive to the available research envelope; failure at twelve batches cannot automatically be attributed to reasoning alone.

PA supplies a particularly strong positive result. Organic-fraction MAE falls from 0.08291 to 0.02379 (13/15 pairs improve), and process-choice accuracy rises from 25/30 to 30/30. Nominal 90% coverage increases from 78.3% to 98.9% while width narrows from 0.25122 to 0.18553. High coverage exceeds its nominal target and is not evidence of perfect calibration. In a selected aligned campaign, coverage of material pairings increases from ten of sixteen to all sixteen, followed by eight process experiments. Two other aligned world pairs worsen, so this successful trajectory is not a universal account of the budget effect.

C improves mean recovery MAE from 0.10433 to 0.07937 but retains 12/15 quality-eligible recommendations at each budget. Retested recovery falls from 0.42054 to 0.39939. Fines coverage remains below nominal in every campaign, rising only from 35.6% to 43.9% overall. Furthermore, the small aggregate size-error improvement reverses when budget pairs containing the nonconforming source are omitted. Full response tables and paired sensitivities accompany the supplementary budget figure. The result is selective resource benefit, not evidence that more experiments cannot help or that data insufficiency has been ruled out.

5 PRIOR VALUE DEPENDS ON THE REGIME AND RESPONSE

Aligned information can help. In RX/P, discovery macro MAE falls from 0.10186 under Opaque to 0.05701 under Aligned; optimization error falls from 0.13893 to 0.08384. Aligned beats Opaque in four of five worlds under either goal. EC also changes ordering with its commission: MisIndexed has the lowest mean score error under discovery at both budgets, while Aligned has the lowest under optimization. Final rankings do not reveal whether a dossier claim was accepted, rejected, or revised.

EQ/P shows a consistent aggregate disadvantage: Opaque macro MAE is 0.01214, versus 0.04314 for Aligned and 0.03928 for MisIndexed, beating both in every world. Aligned supplies a true effective-pKa interval, not the complete coupled law. A post hoc partition of questions by public concentration reveals a more informative reversal. Aligned **improves on the other nine questions but worsens on the three most dilute in all five worlds** (Figure 3). MAEs change from 0.01008 to 0.00465 on the other nine, and from 0.01833 to 0.15862 on the dilute three. The other nine include boundary questions and cannot all be called in-distribution tests.

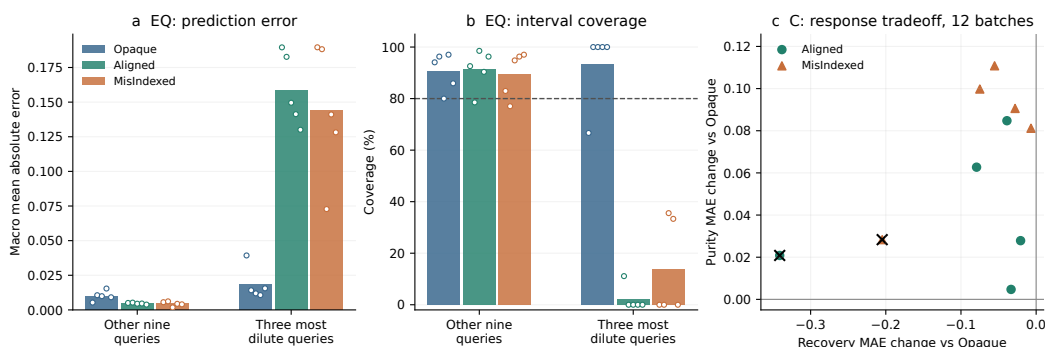


Figure 3: Conditional information effects. EQ/P: mean macro error and interval coverage for the three most dilute recipes and the other nine, with nominal 80% coverage marked. C/12: dossier-minus-Opaque changes in recovery and purity error for every world. Negative recovery and positive purity changes show a benefit and cost in the same pair; crosses include the nonconforming source. The query grouping and joint-response contrast are exploratory, with no independent-query significance test.

Aligned coverage is 91.3% on the other nine questions but 2.2% on the dilute three; Opaque obtains 90.7% and 93.3%. The source campaigns all remain above 0.018 M and observe a dissociation plateau near 0.06–0.09. The dilute questions instead reach approximately 10^{-4} to 10^{-5} M. One aligned campaign spans a ninety-fold concentration range while Opaque spans only fourfold, but neither leaves the source response regime. On a selected dilute query, true dissociation is 0.65463: Opaque predicts 0.629 [0.45, 0.80], Aligned 0.0804 [0.053, 0.108]. Input-range breadth thus need not measure coverage of distinct response regimes. This pattern supports a behavioural diagnosis; it does not identify a causal anchoring mechanism.

C gives a response-level reversal. At twelve batches, both Aligned and MisIndexed improve recovery error while worsening purity error relative to Opaque in all five worlds, including all four both-conforming pairs for each comparison. At 24 batches the joint direction holds in only two and three worlds, respectively. No single arm ranking describes both responses and budgets. Across EQ/E, EQ/S, PA, and RX/S, mixed directions and tail errors further caution against aggregating all priors

into a common treatment effect. Different locus studies also change their specific scientific questions, so comparisons between P and S are not isolated effects of prior type.

6 WHEN SHOULD AN OBSERVED RELATIONSHIP BE PRESERVED?

To connect prediction to acquired evidence, we compare every C campaign with two empirical references fitted to its public final observations: the response mean and nearest-neighbour prediction from public recipe features. They obtain no additional experiments or hidden truth during fitting. They are simple references, not strong system-identification baselines or replicas of the agent’s full information and computational policy.

Agent recovery predictions beat the public mean in **26/30** campaigns. Purity predictions lose in **30/30**, and size predictions in **26/30** (Figure 4). Mean purity error is 0.05029 and 0.03679 at the two budgets, compared with 0.00475 and 0.00489 for the observation mean. Source and target purity are both approximately 0.985, while predicted means are 0.93544 and 0.94893. In total, 335/360 query predictions are too low. The low within-world target span, about 0.02442, explains why the constant reference is strong; its success does not establish mechanism discovery.

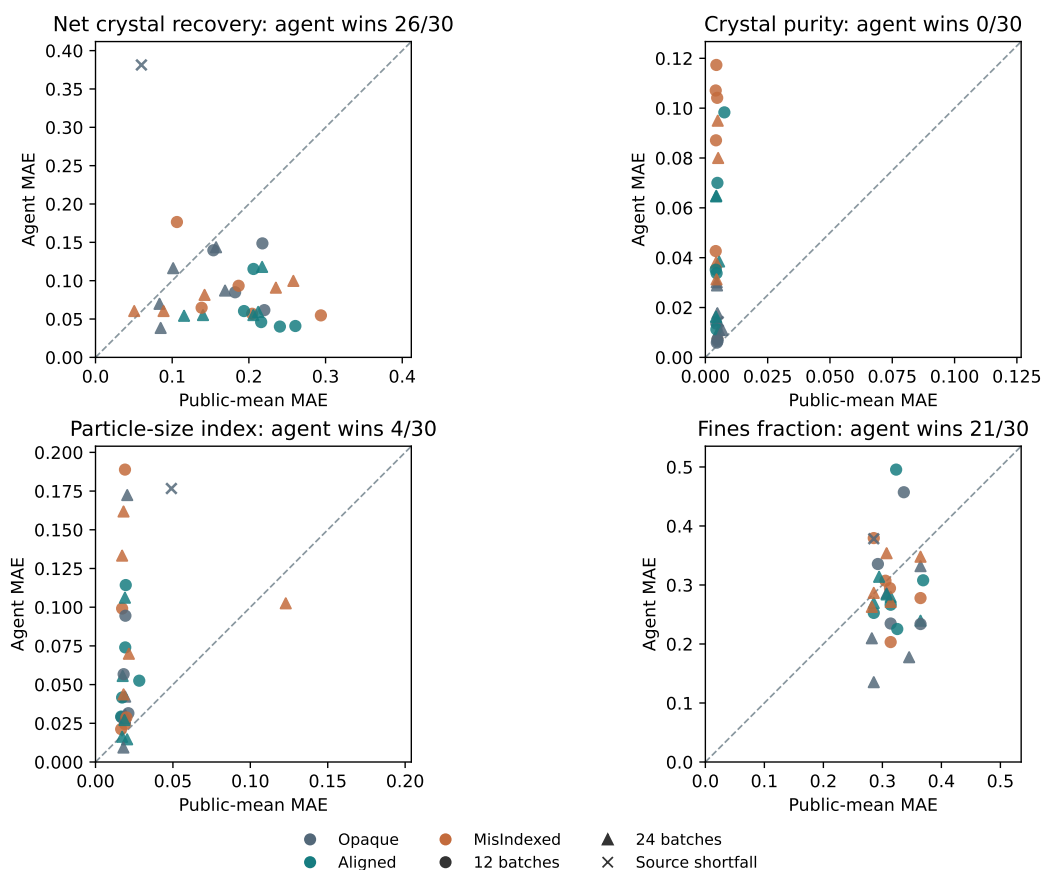


Figure 4: C forecasts versus the mean of their own source observations. Above-diagonal points favour the public mean. The contrast is selective across responses: net recovery generally improves, while purity always worsens. Circles and triangles denote budgets; the cross marks the eleven-assay source. Each panel has its own scale and reuses five worlds.

A selected misindexed C campaign explicitly acknowledges in K1 that it tested only one solvent. Yet its purity predictions average 0.86833 against 0.98566 in the references, and none of twelve nominal 80% intervals covers the target. The example, selected for high purity error, shows a possible disconnect between stated limitations and quantified uncertainty; it does not estimate its prevalence.

Likewise, selected EC traces link restricted potential coverage to poor predictions in the untested regime, but counterexamples show that sparse sampling alone does not determine failure.

EQ and C require opposite uses of empirical stability. An EQ source-mean diagnostic fails badly on dilute questions because the plateau should break; the C mean is effective because purity remains stable on its test distribution. The common evaluation question is whether an agent can identify **when to retain a relationship and when to revise it**. This synthesis is a testable interpretation of the observed patterns, not a demonstrated single cause. Predictions retain full context, so none of these comparisons establishes information loss through report compression.

7 RELATED WORK, LIMITATIONS, AND IMPLICATIONS

Chemistry agents and autonomous laboratories demonstrate planning and execution (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023). Summit and Olympus support repeatable optimization studies (Felton et al., 2021; Häse et al., 2021); ChemGymRL supports interactive chemistry (Beeler et al., 2024). DiscoveryWorld and the BoxingGym preprint study scientific experimentation in controlled environments (Jansen et al., 2024; Gandhi et al., 2025). ChemWorld’s particular contribution is to combine composable processes, persistent material and observation histories, and matched information interventions with separate measurements of task achievement and prediction. The distinction between decision quality and prediction error also connects to decision-focused learning (Elmachtoub & Grigas, 2022).

The evidence is limited to one model configuration and five worlds per study, with correlated readouts and no repeated independent model seeds within a cell. There is no wet-laboratory validation. The EQ grouping and trace examples are post hoc; the C baseline exploits a low-variation response. Early EC/PA calculator limits and retained RX/EQ prompt language differ from later blocks. These differences constrain cross-system interpretation. Complete mechanism annotation, strong classical sequential comparators, and causal evidence-matched interventions remain open work, rather than implied results.

The findings motivate evaluation along three linked axes: what the agent achieves, what it predicts under new interventions, and whether its uncertainty matches its evidence. Improvements on one axis should not substitute for measurements on the others. ChemWorld provides the controls and records needed to study their relationship while preserving autonomous scientific behaviour.

REPRODUCIBILITY STATEMENT

Appendix A specifies the full study matrix, targets, resources, failures, and assessment chronology. Appendix B reports every response-specific arm mean; Appendix C details the exploratory subgroup and joint-response analyses. Source campaigns, failed attempts, public observations, and evaluation records are retained for reconstruction. The anonymous code and campaign-level data package for this integrated study is still being assembled; this draft does not claim that a complete supplementary code release has already been supplied.

AI USE STATEMENT

Generative AI assisted software implementation, experiment orchestration, retained-data analysis, visualization, and manuscript drafting. Agent outputs are experimental observations rather than independent validation of the paper’s claims. Responsibility for scientific interpretation and the submitted manuscript remains with the authors.

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A PROTOCOL DETAIL AND EVIDENCE HANDLING

A.1 COMPLETE STUDY MATRIX

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		240	3,597 / 3,600

Table A1. Complete selected study scope. Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have sealed mechanism, prediction, and reflection stages; 238 meet the full source-and-assessment specification.

A.2 PREDICTION TARGETS AND SYSTEM-SPECIFIC QUESTIONS

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

EC's twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX's twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX's 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. EQ/E crosses three medium identities with three concentrations and a matched-concentration scale control, assessing entity contrast and concentration and scale transfer under a fixed common topology. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument's equilibrium diagnostic is not the agent's subjective confidence. The bounded study's effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C's twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250 \mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P's six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C's crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system's responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.3 LANGUAGE, NUMERICAL TOOLS, AND STAGE BUDGETS

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.4 COMPLETION, RECOVERY, AND EXCLUSIONS

The primary denominator is 240 scheduled campaigns, not provider attempts. All 720 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,597 of 3,600 planned. C's second-world 12-batch Opaque source has eleven final assays. P's fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported in the main text. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 240 independent chemistry replications.

A.5 BASELINE CORRECTION AND TRACE INTERPRETATION

The C empirical references use the same source final observations for all three arms. A recipe parser originally rejected resource-denial transaction records in two sources. The correction preserves those attempts as provenance and excludes them from committed physical recipe features. Recomputing all thirty baselines restores the two missing evaluations and exactly reproduces the twenty-eight previously available results. Original trajectories, truth, model answers, and error records are retained; the correction uses no new simulator or provider calls. Unknown transaction statuses remain errors. This is an evaluator repair, not a change to a completed experiment.

The EC illustrative trace is the second-world, 12-batch Opaque EC discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. They are retrospectively selected examples, not a coded prevalence estimate. The

594 additional C case is the fifth-world, 12-batch MisIndexed campaign, chosen as the largest purity MAE
595 in the 12-batch cohort. Its extreme error illustrates a failure; it is not a representative sampling rule.
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597 A systematic analysis should attach each mechanistic claim to its experimental support, contradictory
598 observations, stated domain, and corresponding predictions. It should distinguish an unperformed dis-
599 criminating experiment, acquired evidence omitted from an account, justified unresolved alternatives,
600 and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions;
601 a claim is not wrong merely for using an equivalent representation. Independent review can assist this
602 analysis without replacing measured predictions and retests. No completed cohort-wide mechanism
603 score or causal compression experiment is implied.
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A.6 FULL PRIOR OVERVIEW

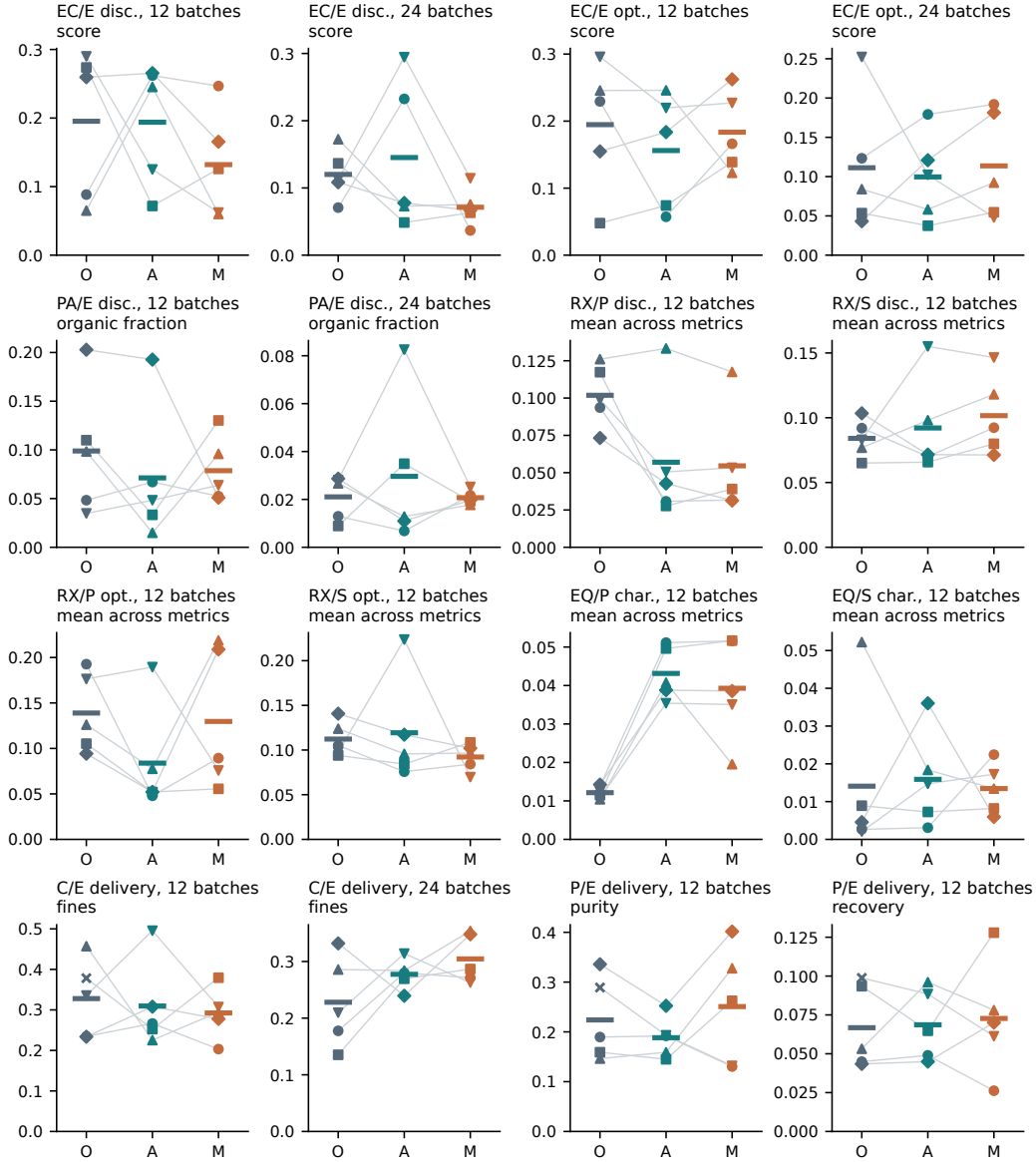


Figure 5: Prior information has no common ordering. Every panel includes five worlds and three arms: Opaque (O), Aligned (A), and MisIndexed (M). Gray lines connect worlds; colored bars show means. Crosses mark source shortfalls. RX and EQ use within-system macro errors; other panels name their response. Scales differ. Appendix B complements these selected readouts with all metric-specific arm means, including EQ/E.

A.7 BUDGET RESPONSE OVERVIEW

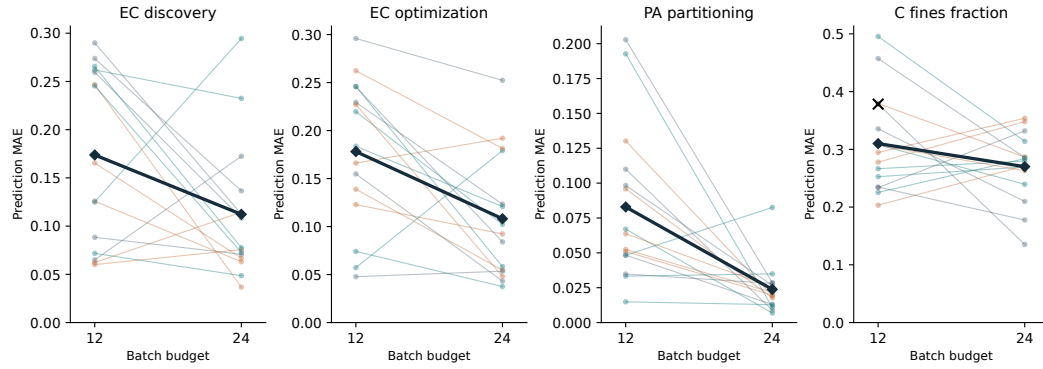


Figure 6: Matched budget effects across responses. Each larger-budget source is a new session with a larger research envelope. World-level pairs and response-specific scales are retained.

B COMPLETE METRIC-SPECIFIC ARM MEANS

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign’s interval width and protocol status.

B.1 EC, E PRIOR: DISCOVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

B.2 EC, E PRIOR: OPTIMIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

B.3 PA, E PRIOR: DISCOVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

B.4 EC, E PRIOR: DISCOVERY, 24 BATCHES

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

B.5 EC, E PRIOR: OPTIMIZATION, 24 BATCHES

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

B.6 PA, E PRIOR: DISCOVERY, 24 BATCHES

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

B.7 RX, P PRIOR: DISCOVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

B.8 RX, P PRIOR: OPTIMIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

B.9 RX, S PRIOR: DISCOVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

B.10 RX, S PRIOR: OPTIMIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

B.11 EQ, P PRIOR: CHARACTERIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

B.12 EQ, S PRIOR: CHARACTERIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

B.13 EQ, E PRIOR: CHARACTERIZATION, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

B.14 C, E PRIOR: DELIVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

B.15 C, E PRIOR: DELIVERY, 24 BATCHES

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

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B.16 P, E PRIOR: DELIVERY, 12 BATCHES

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

C EXPLORATORY ANALYSES OF RESPONSE REGIMES

These analyses use the retained source campaigns and their original predictions. They do not define new experimental arms or alter the withheld-condition bank. The concentration grouping, joint crystallization contrast and illustrative traces were selected after the experiments, and are descriptive analyses of reused worlds.

C.1 EQUILIBRIUM: A REVERSAL ACROSS PUBLIC CONCENTRATION REGIMES

The three most dilute recipes have nominal concentrations approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Each campaign has three normalized responses for each recipe. Point errors use five-observation means; interval coverage uses the individual reference observations. Group means average the corresponding responses and queries and then the five world campaigns. Weighting the two groups 3:9 reconstructs every original twelve-question campaign macro MAE. The other nine queries include boundary conditions; this split is not a predeclared interpolation-versus-extrapolation test.

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Aligned improves on the other nine and worsens on the dilute three relative to Opaque in all five worlds. All fifteen source campaigns remain above 0.018 M. In the selected second-world comparison, Aligned spans about ninety-fold source concentration variation and Opaque about fourfold, but both remain in the observed plateau. At the most dilute question, reference dissociation is 0.65463; Opaque predicts 0.629 [0.45, 0.80] and Aligned 0.0804 [0.053, 0.108]. The Opaque prediction response develops an unsaturated weak-acid extrapolation beyond its sealed report. This reasoning must not be credited retrospectively to K1 or to an earlier source-stage hypothesis.

C.2 CRYSTALLIZATION: JOINT RESPONSE CHANGES UNDER THE SAME TREATMENT

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

C.3 EVIDENCE STATUS AND INTERPRETATION

The same-source C references use only each campaign’s final assays and public recipe features. They exclude hidden truth during fitting but do not reproduce the agent’s complete intermediate measurements, prior or computational policy. Low held-out purity variation makes the mean a particularly strong reference. Neither its success nor the agent’s recovery advantage is a complete mechanism identification result.

Selected EC and C traces support detailed examples rather than population frequencies; the bounded EQ review below covers all fifteen campaigns in that block. The current evidence does not causally identify prior anchoring, report compression, acquisition-versus-inference mediation or a general preference for familiar chemistry. Full-history Q also prevents interpreting K1 as the sole information channel. Experimental selection and later interpretation are both part of the original autonomous process. Their separate causal contributions are not required to describe the observed whole-process contrasts and have not been identified here.

C.4 ORIGINAL EQUILIBRIUM CAMPAIGNS: PREDICTIONS AND RESEARCH CHRONOLOGY

We reviewed all fifteen retained parameter-prior equilibrium campaigns, comprising 180 batches, 945 operations and 45 K1/Q/K2 stages. Each row below is an original autonomous research session, not an independent reader of its records. The selected question uses 1 micromole in 75 mL; the same question and original reference targets apply to all three arms in a world. The reference column is the five-observation mean.

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

The first-world Opaque K1 already estimates an effective acid constant near 2.2×10^{-5} and an active-pool cap near 0.004 M, while retaining competing interpretations. Its Q applies that hypothesis at low concentration. In the second world, Opaque K1 primarily supports the observed plateau and limits extrapolation; the unsaturated weak-acid calculation is first explicit in Q. The aligned K1 similarly limits its account to the sampled region, but its Q interval still misses the dilute response by a large margin. These differences separate what was stated before prediction from what became explicit during it.

All fifteen source concentration ranges lie above the three dilute test recipes. The exported per-operation decision-reason fields are unavailable for all 945 operations. Actions and observations can therefore be reconstructed, but a contemporaneous sequence of belief revisions cannot be inferred from these fields. K1, Q and K2 remain distinct public records rather than substitutes for that sequence.

In K2, eleven of fifteen campaigns propose a trace-load experiment: five Opaque, three Aligned and three MisIndexed. The other choices are equal-concentration scale-up, dilution near the supplied relation, comparison of instruments at the same state and repetition of an anomalous batch. K2 follows exposure to Q conditions without feedback on their reference outcomes. None of these proposals was executed; they are retrospective choices, not source-stage plans or demonstrated repairs.